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Sound of freshness: Crafting multisensory experience in perfumery

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ABSTRACT

The current landscape of the perfume industry faces the challenge of enhancing product appeal and captivating consumers through innovation. Integrating multisensory elements into perfume experiences has the potential to drive this innovation forward. This research presents a novel methodology for characterizing sounds that evoke the perception of freshness in fragrances. Three experiments were conducted to explore the transferability of identified sound characteristics into the experience of certain fragrances. Experiment 1 assessed selected sounds to identify auditory parameters that effectively triggered primary freshness attributes. Based on these results, originally composed brand-aligned soundtracks were tested in Experiment 2 using the Implicit Association Test (IAT) to investigate associations with freshness and brand values. Experiment 3 examined whether these soundtracks effectively modulated the implicit olfactory experience of two commercial fragrances. The findings showed that the soundtracks successfully elicited specific enhanced effects in how people implicitly perceived the freshness of a fragrance. These outcomes revealed how soundtracks can enhance the perceptions of freshness attributes (e.g., cold, blue, light) in the fragrance experience, consistencies and variations between self-report and implicit measures, and practical applications in multisensory strategies for cosmetic companies.

1. Introduction

In the ever-evolving landscape of industry and organizations, staying attuned to shifting consumer needs and desires is crucial. By adopting a consumer-centric approach, products and experiences can be curated to incorporate sensations and emotions, fostering a deeper connection between consumers and brands across offline and online channels. Experiences enhance people's emotions and product/service attractiveness, creating strong positive impressions (Eglīte, 2021). The rationale of crossmodal sensory associations applied to the perceptions related to a product's experience, or the environment that surrounds such an experience (Bourdier et al., 2023), allows for the delivery of structured multisensory experiences and a more humanistic and enchanting approach that adds substantial consumer value (Eglīte, 2021).

The perfume industry has the opportunity to actively engage in this discussion to continue innovating while delivering the positive emotional impact sought by consumers. The sense of smell plays a significant role in the overall consumer experience of cosmetics, evoking emotionally charged and vivid encounters unlike other sensory modalities (Herz & Cupchik, 1995). For instance, a recent study demonstrated that sensory multimodal fragrance–texture congruence results in combined hedonic response positively influencing overall cosmetic product appreciation (Bourdier et al., 2023).

To this end, and from a novel perspective, auditory cues as part of perfume experiences may be helpful. Seo and Hummel (2011) demonstrated that the hedonic valence associated with sounds could be transferred to aromas (as in sensation transference; Spence et al., 2021). Seo et al. (2014) also reported on a series of three experiments in which people matched various aromas to background music, revealing that congruent background music enhanced the rated pleasantness of certain scents. Here, congruent sounds also influenced participants' familiarity and identification of the aromas.

Previous studies have approached the sound-scent relationship from different perspectives. For example, people with vision impairment

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successfully associated colors and sounds with specific scents (Quero et al., 2021), while food-related aromas were associated with musical features (e.g., Crisinel & Spence, 2010, 2012; Speed et al., 2021). Seo et al. (2014) further explored this relationship using urban sounds, finding associations with lilac, osmanthus, coffee, and bread aromas.

Motivated by the existing findings, the present work introduces an innovative methodology for characterizing sounds that crossmodally correspond to particular sensations related to the experience of fragrances. This methodology identifies semantic, sensory, and emotional descriptors of this target experience across different modalities (e.g., temperature, emotional attributes, colors) and pairs them with musical characteristics (e.g., pitch, instruments, genre). This results in soundtracks comprising these characteristics potentially modulating the fragrance experience of perfumes.

In the design of this study, the focus lied on investigating particularly the olfactory experience of freshness. Fresh is commonly classified among the main families of perfumery, alongside oriental, woody, and floral (Edwards, 1984), and it is the most recurring scent reported by Thiboud (1994). However, the interpretation of freshness in olfactory description lacks consensus, which may explain why some perfumery companies do not consider it as a central family (Zarzo, 2012). Additionally, the multidimensional complexity of olfactory perception makes it difficult to measure and describe (Wise et al., 2000), with commercial perfumes, created by blending odorous materials with varying volatilities, introducing greater complexity (Zarzo, 2012). Hence, investigating the role of sound in the experience of freshness in perfumery may provide novel contextual cues for enhancing perfumery experiences. While prior research had touched upon people's associations between fresh fragrances and natural sounds (Mahdavi et al., 2020), to the best of the authors' knowledge, no studies had explored the role of sound in the experience of freshness in perfumery, making this investigation exploratory in nature.

1.1. Literature review

1.1.1. Disentangling the experience of freshness

The study of freshness in personal care is limited compared to, for example, foods and beverages (e.g., Roque et al., 2018). Nevertheless, an example outside the food industry is around dishwashing liquids, in which peppermint, bergamot lemon aromas, and light green color were considered fresher stimuli, while patchouli, almond aromas, and purple/dark brown color were seen as the least fresh (Fenko et al., 2009). Importantly, olfaction showed a stronger influence on freshness judgments than visual cues (i.e., color).

In cosmetic products, aroma stimuli, temperature change, oily residue, spreadability, and the ability to penetrate the skin play a key role in the perception of freshness by consumers (Damonte et al., 2011; Vieira et al., 2020). Research in perfumery has focused on identifying aroma descriptors associated with the concept of freshness, including citrusy, clear, leafy, minty, watery, floral-green, blue, fruity, light, and aldehyde notes, as well as materials with low substantivity (the durability of material when applied to the skin; Zarzo, 2012). Conversely, powdery, high sweet, erogenic, animal, and oriental are descriptors associated with opposite sensations to freshness. These findings are attributed to the fact that aromas with higher substantivity tend to be more pronounced in hot weather due to their stronger adherence to the skin.

Consumers associate multiple semantic and sensory parameters with the experience of freshness across diverse senses. While research on the association of fresh fragrances with sounds is scarce, some designers have linked fresh fragrances to specific musical tones. For example, fragrances classified as fresh were mapped to mid-high pitch piano keys based on an ancient classification (C.-H. Lee, 2013). More recently, researchers found that the sounds of nature, such as the sea and rain, were associated with a fresh perception of fragrances (Mahdavi et al., 2020).

1.1.2. Translating fresh descriptors into sound parameters

Low temperatures (0–10 °C) are associated with low arousal and mid-low valence emotions, including blue/uninspired, dull/bored, and passive/quiet states (Barbosa Escobar et al., 2021). Similarly, winter visual scenery in virtual reality simulations has been linked to perceptions of mint, moderate wind, and cooling sensations (Ranasinghe et al., 2018). Chamomile-roman aroma and blue color are commonly associated with low arousal (Fixsen, 2019; Fukumoto, 2020).

Previous studies have demonstrated that piano sounds effectively represent the color blue, while woodwind instruments like flute, clarinet, and bassoon evoke associations with green (Bologna et al., 2007; Cho et al., 2020). Additionally, light and cool color variants correspond to lemon and caramel scents, respectively (Quero et al., 2021). Vanilla aroma is connected to low temperatures, thick silhouettes, pleasantness, low intensity, and round shapes (Brianza et al., 2021; Hanson-Vaux et al., 2013; Jezler et al., 2016). Mid-high-pitched notes consistently match vanilla, violet, and blackberry aromas, while high pitches match lemon and candied orange. Blackberry, apricot, raspberry, vanilla, and candied orange aromas often match piano and woodwinds (Crisinel et al., 2013; Crisinel & Spence, 2010, 2012). Fruity, soft/light, and cool scents are associated with nature sounds (e.g., sea, rain, waves, wind, breeze) and calm/soft music (Mahdavi et al., 2020). Whereas lemon smell is typically classified as fresh, previous studies have linked it to non-fresh concepts, such as summer visual cues (Ranasinghe et al., 2018) and high-intense smell (Hanson-Vaux et al., 2013).

Table 1 provides a summary of established associations related to temperature (low), emotions (low arousal), colors (green, blue), flavors (peppermint), and smells (fruity, lemon, mint, cool, light) that consistently align with the sensory experience of freshness. These features can also be translated into sound parameters, including timbres (woodwind, piano), tempo (low), pitch (mid-high), mood (calm, soft), and naturalistic soundscaping (nature, breeze, sea, wind, rain).

1.2. The present study

Inspired by the above literature review, this study aimed to identify the conveyance ability of sensations associated with the experience of freshness in fragrances and evaluated this conveyance in the context of a perfume brand. Three experiments were conducted to achieve this goal (Fig. 1).

Experiment 1 involved an online assessment of baseline instrumental sounds previously validated as resembling fresh or non-fresh sensations. For Experiments 2 and 3, original brand-aligned soundtracks were composed based on the results of Experiment 1.

Experiment 2 used the Implicit Association Test (IAT) to explore unconscious associations between these soundtracks and attributes of freshness and brand archetype. The Implicit Association Test (IAT) is a tool developed by Anthony Greenwald et al. (1998) to principally measure individuals' automatic associations between mental representations of concepts, aligning well with this study's goals. Such a test involves quickly categorizing stimuli (e.g., sounds, words, images, scents, and even tactile), revealing explicit/implicit biases that may exist unconsciously. Despite some controversy surrounding its reliability, the IAT continues to be a valuable instrument for psychology research and highlighting implicit associations, offering insights into attitudes that may influence behavior.

Experiment 3 assessed in a laboratory setting whether the brand-aligned soundtracks validated in Experiment 2 were able to modulate, in an enhanced manner, the olfactory experience of two commercial perfumes, using the IAT.

Seeking to enhance the consumer experience through multisensory engagement, a major Brazilian cosmetics company attempted to amplify the perception of freshness in one of its fragrance lines. This attempt involved the development of a soundtrack that would enhance specific freshness-related attributes for this line's brand. The brand values predominantly featured blue palettes and explorer/outdoorsy/energetic

Table 1
Sensory and emotional associations that people consistently match with the sensory experience of freshness.

Freshness dimension	The sensation associated with freshness	Chemosensory sensation crossmodally corresponding	Visual sensation crossmodally corresponding	Semantic context crossmodally corresponding	Auditory sensation crossmodally corresponding	References
Temperature	Low temperature			Low arousal		Barbosa Escobar et al. (2021)
Emotional	Low arousal	Roman chamomile			Low tempo	Fukumoto (2020)
Visual	Green color				Clarinet, bassoon, and flute	Cho et al. (2020)
	Blue color	Musk and sea salt scent and blue-berry flavor		Calm and dependability	Piano	Cho et al. (2020); Fixsen (2019)
Flavor	Peppermint flavor				Piano and high pitch	Crisinel & Spence (2010)
Olfaction	Fruity smell	Raspberry and vanilla	Rounded shapes and thick silhouettes	Pleasantness, low intense, and coldness	Piano, woodwind, high-pitch, mid-pitch, and nature sounds	Brianza et al. (2021); Crisinel et al. (2013); Crisinel & Spence (2010; 2012); Hanson-Vaux et al. (2013); Jezler et al. (2016); Mahdavi et al. (2020)
	Lemon smell		Spiky shapes and thin body silhouettes	Summer, unpleasantness, intense, and feminine	Woodwind	Crisinel & Spence (2010); Hanson-Vaux et al. (2013); Jezler et al. (2016); Ranasinghe et al. (2018)
	Cooling mint scent	Eucalyptus and peppermint	Spiky shapes and thin body silhouettes	Winter, coldness, and high arousal	High pitch	Brianza et al. (2021); Ranasinghe et al. (2018)
	Cool scent				Breeze, sea, wind, and rain sounds	Mahdavi et al. (2020)
	Light scent				Sea, rain, calm, and soft sounds	

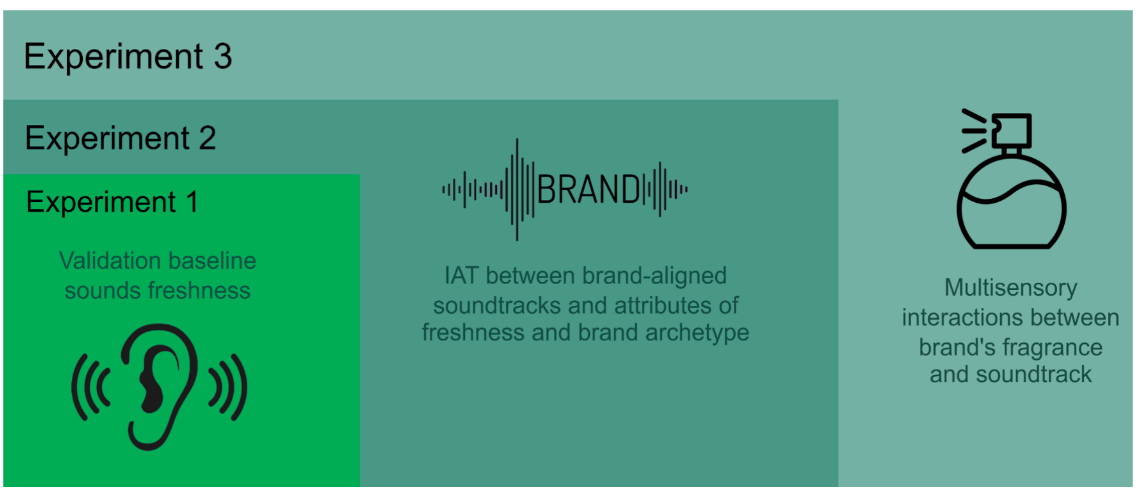

Fig. 1. Illustration of the stimuli used in the experiments of this study. Experiment 1 involved baseline sounds, Experiment 2 added a brand-identity layer to the sounds, and Experiment 3 added fragrance stimuli.

experiences, inviting users to explore humid cold natural scenes (i.e., moist and cold environments), while feeling light and calm. Hence, the commercial perfumes used in the study (particularly, in Experiment 3) were selected considering such a brand strategy.

The olfactory profile of the experimental fragrances in this study was guided by the outstanding brand archetype of watery, energetic, natural, and cleanliness, aligned with the brand values. In fragrance development, the *fougère* family, recognized as the aromatic family, stands as the primary representative in terms of olfactory freshness. Within this family, two distinct olfactory aspects of *fougère* freshness are commonly identified—namely, mountain and forest. The experimental fragrances in this study embodied the mountain freshness, characterized by a colder, icy, cool, watery, and refreshing connotation, achieved through ingredients such as dihydromyrcenol found in citrus peel.

Importantly, the target consumers of this company implied that the data collection for all three experiments involved exclusively Brazilian participants and used the Portuguese language.

2. Experiment 1

Experiment 1 assessed baseline sounds across sensations, fragrance descriptors related to fresh experiences, and color matching. Online participants were asked to listen to these sounds and rate them using rating scales. The analysis focused on identifying the musical features of the sounds that evoked the strongest sensations of freshness.

2.1. Methodology

2.1.1. Participants

A sample of 794 Brazilian participants joined Experiment 1 (44 % female), with all of them over 18 years old (50 % aged 21–35, 40 % aged 36–45). Netquest, a data collection provider (https://www.netquest.com/), was used for recruitment. All participants were rewarded with points that could be exchanged for money or other benefits on services/platforms. They were required to reside in Brazil, with 80 % being frequent consumers/shoppers of the Brazilian company perfumery, and

34 % specifically using the target brand.

An a priori power analysis was conducted using G*Power version 3.1.9.7 (Faul et al., 2007) for sample size estimation, based on data from Cantu et al. (2022) (N = 376), who assessed the influence of sounds on sensations. The effect size in Cantu et al.'s study was 0.15, considered small using Cohen's (1988) criteria. With a significance criterion of α = 0.05 and power = 0.99, the minimum sample size needed with this effect size is 696 for multivariate ANOVA. Thus, the obtained sample size of 794 is sufficient to assess the effect of sound on sensations.

2.1.2. Materials and stimuli

This experiment was based on two online experiments presented at the IFSCC 2022 (Cantu et al., 2022), which provided most of the sounds described below. Six sounds showing general evidence of triggering sensations of freshness and two eliciting contrasting sensations were selected for Experiment 1. The fresh sounds (F1 to F6) featured woodwind and piano instrumentation, low tempo, mid-high pitch, legato articulation, and conveyed a calm/soft mood, evoking low-arousal emotions (see Table 1). Some of these sounds also incorporated Autonomous Sensory Meridian Response features (Barratt et al., 2017). The contrasting sounds (NF1 and NF2) represented genres associated with tropical/warm areas, featuring a high tempo, strings and/or brass instrumentation, and staccato articulation, evoking high-arousal emotions and sensuality (cf. Table 1). All eight auditory stimuli used in Experiment 1 lasted between 45 and 60 s and can be accessed at https://osf.io/va2sz/.

2.1.3. Measures

Experiment 1 assessed three categories of dependent variables: sensations, olfactory notes, and color. Based on the existing framework, ten paired sensations (hot–cold, dry-humid, urban-outdoor, unpleasant-pleasant, uncomfortable-comfortable, sweaty-cool, energetic-calm, powdery-smooth, heavy-light, cautious-explorer), representing environmental, emotional, physical, and lifestyle dimensions, were evaluated using 7-point bipolar scales. Each scale was anchored with "very much," "enough," and "somewhat" on each side, and "balanced" in the middle.

For olfactory notes, participants indicated how the sound they heard evoked six fragrance descriptors. Three descriptors (fresh, fruity, and citrus) represented freshness, while two (spicy and spices) represented contrasting sensations. An additional descriptor (sweet) was related to freshness at a medium intensity. These descriptors were evaluated on 7-point unipolar scales, from "not at all" to "very much." Selected olfactory references (e.g., mint, strawberry, peppers) facilitated participants' associations with previous experiences (Herz & Cupchik, 1995).

Finally, the color dimension examined participants' associations between sounds and colors. Participants selected one color among primary and secondary (yellow, orange, red, purple, blue, green; Wright, 2018) that they felt best matched what they heard.

2.1.4. Experimental design and procedure

Experiment 1 randomly assigned subjects to one out of eight auditory conditions (six fresh and two non-fresh), creating eight between-participants conditions. The survey was administered online through Qualtrics (https://www.qualtrics.com), with an estimated duration of four minutes. Participants provided informed consent and were instructed to always use headphones. Prior to the main experiment, participants completed an auditory acuity test to ensure proper usage.

During the experiment, participants randomly listened to one of the available sounds and were asked to attentively think about it. While listening, they rated ten environmental, emotional, physical, and lifestyle sensations on bipolar scales. Participants also rated the extent to which the sound evoked six olfactory notes and were asked to select a corresponding color (section 2.1.3). Age, gender, occupation, and perfumery consumption/choice were collected at the end of the survey.

2.1.5. Data analysis

Statistical analyses used IBM SPSS 26.0 (IBM Corp.; Armonk, NY, USA). Multivariate ANOVA general linear models assessed main effects on sensation scores and olfactory notes, independently, with sound as a fixed factor. Age and gender were included as covariates. Pairwise comparisons were Bonferroni-corrected. Pearson's method determined correlation scores for covariate effects. Moreover, a Principal Component Analysis (PCA) visually represented the differences between fresh and non-fresh sounds.

2.2. Results

The results of Experiment 1 showed that fresh sounds evoked significantly higher scores on cold ($F(7, 780) = 29.19$; $p < 0.001$; $\eta^2 = 0.21$), outdoor ($F(7, 780) = 35.88$; $p < 0.001$; $\eta^2 = 0.24$), cool ($F(7, 780) = 36.09$; $p < 0.001$; $\eta^2 = 0.25$), and calm ($F(7, 780) = 61.48$; $p < 0.001$; $\eta^2 = 0.36$) sensations than the non-fresh ones (Fig. 2). Most fresh sounds also elicited significantly higher scores on light sensation ($F(7, 780) = 20.38$; $p < 0.001$; $\eta^2 = 0.16$), whereas the non-fresh ones elicited higher scores on explorer attribute ($F(7, 780) = 13.28$; $p < 0.001$; $\eta^2 = 0.11$). Among the fresh sounds, F1 was associated with humid, outdoor, and cool, whereas F2 was associated with hot, dry, pleasant, and smooth sensations (see Table A1 for details).

In terms of olfactory notes, most fresh sounds evoked significantly lower scores on the spicy note ($F(7, 780) = 61.04$; $p < 0.001$; $\eta^2 = 0.35$) and higher scores on the fresh note ($F(7, 780) = 15.70$; $p < 0.001$; $\eta^2 = 0.12$) than the non-fresh sounds (Fig. 3). F2 was associated with fruity and sweet notes, whereas NF2 was associated with spices. Citrus scores did not differ significantly between the sounds (see Table A2 for details).

Regarding color evaluation, green and blue were the preferred associations for fresh sounds, while non-fresh sounds were associated with red color (Table 2).

Table 3 summarizes the main sound associations in Experiment 1. F1, F2, F3, and F4 triggered freshness attributes, avoiding the recall for contrasting notes.

3. Experiment 2

Experiment 2 aimed to find subtle associations of brand-aligned soundtracks with freshness and brand archetype attributes. F1 and F3 sounds were selected from Experiment 1 as the most aligned with the brand's desired freshness attributes. These sounds served as a foundation for two new bespoke soundtracks aiming to balance commercial validity while triggering a sense of freshness. Participants in Experiments 2 and 3 were specifically chosen to align with the target user profile of the target brand, comprising males aged 29–49 years old. In Experiment 2, they completed an online IAT across attributes of freshness and brand archetype after listening to the commercial soundtracks or white noise (control).

To move beyond the stereotypical self-report measures, we used the IAT tool (Greenwald et al., 1998). Such an inclusion allowed us to explore beyond explicit perceptions in the sound-freshness relationship, observing not only conscious but also unconscious associations. Although self-report measures are the most common in the field (Rodriguez et al., 2023), they rely on individual introspective abilities, and are more susceptible to biases than IAT (Lemercier-Talbot et al., 2019). In this study, IAT was used to measure the strength of associations between target concepts (sounds or fragrances) and attribute dimensions (e.g., fresh, cold, nature). IAT seems relevant since people tend to respond more quickly/accurately when targets share attributes with the same response key. Hence, we defined that the strength of the associations would depend on the individuals' reaction times.

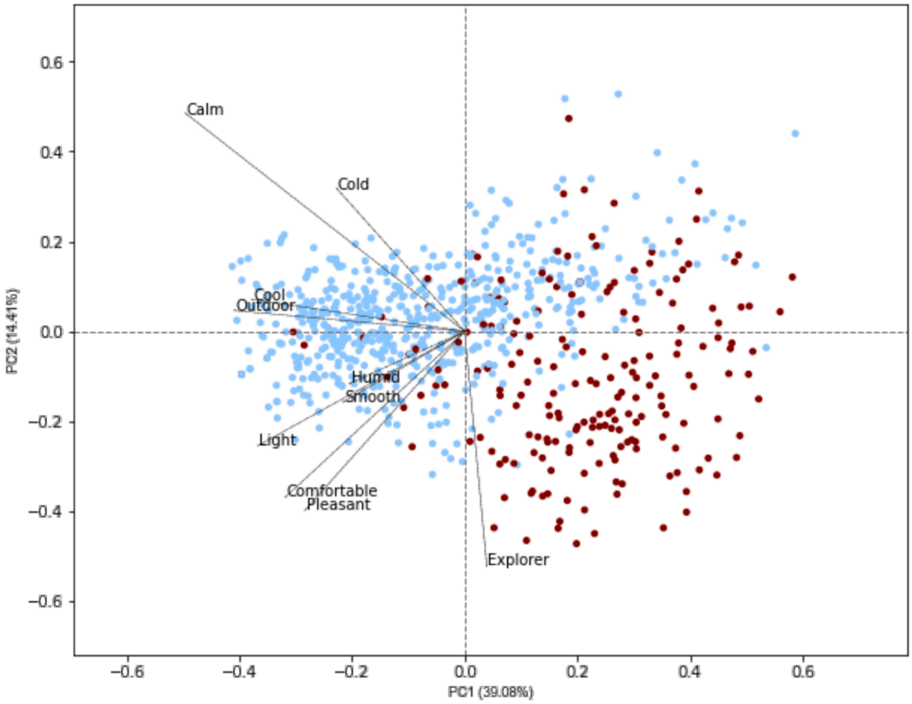

Fig. 2. PCA visualization on the scores for the sensation attributes assessed in Experiment 1. Light blue points represent the scores while listening to the fresh sounds, and dark red points represent the scores while listening to the non-fresh sounds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

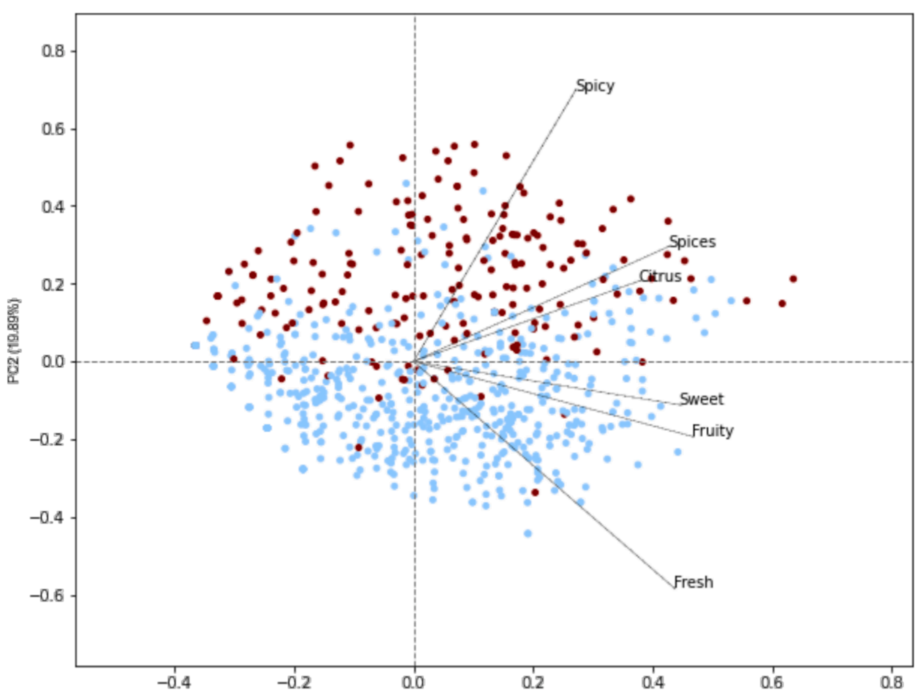

Fig. 3. PCA visualization on the scores for the olfactory note attributes assessed in Experiment 1. Light blue points represent the scores while listening to the fresh sounds, and dark red points represent the scores while listening to the non-fresh sounds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1. *Methodology*

3.1.1. *Participants*

Ninety participants (male, aged 29–49 years, M = 39 years, SD = 5) joined Experiment 2, representing the target brand consumers. All participants were rewarded with points that could be redeemed on e-commerce platforms. They were regular perfume users (80 % of the brand) and residents of Brazil. Recruitment was conducted through Offerwise, a Latin American consumer research participant recruiting agency (https://offerwise.com/).

Table 2
Sum of choices for each color associated with each sound in Experiment 1. The first and second choices for each soundtrack are highlighted in bold.

Sound	Yellow	Orange	Red	Purple	Blue	Green
F1	6	6	3	5	**24**	**61**
F2	18	12	2	13	**23**	**32**
F3	11	16	1	6	**30**	**37**
F4	12	8	2	10	**39**	**25**
F5	12	12	5	16	**30**	**18**
F6	11	19	0	13	**22**	**34**
NF1	3	**26**	**51**	6	3	8
NF2	5	**18**	**55**	9	7	9

Table 3
Significant differences between the sounds in Experiment 1 across sensations, olfactory notes, and colors.

Sound	Sensations	Olfactory notes	Colors
F1	Cold, humid, outdoor, cool, light, calm	Fresh, low spicy	Green, blue
F2	Dry, pleasant, comfortable, smooth, light, calm, cautious	Fresh, fruity, low spicy, sweet	Green, blue
F3	Cold, humid, calm, cautious	Low spices, low spicy	Green, blue
F4	Cold, humid, calm, cautious	Low citrus, low spicy	Blue, green
F5	Dry, heavy, calm, cautious	Low spicy	Blue, green
F6	Dry, calm, cautious	Low fruity, low spicy	Green, blue
NF1	Hot, dry, urban, sweaty, energetic, explorer	Spicy, low fresh, citrus	Red, orange
NF2	Hot, dry, urban, unpleasant, uncomfortable, sweaty, powdery, energetic, explorer, heavy	Spicy, low fresh, spices	Red, orange

An a priori power analysis using G*Power 3.1.9.7 (Faul et al., 2007) determined the minimum sample size. With an effect size of 0.5 (medium size according to Cohen, 1988), $\alpha = 0.05$, and power = 0.99, the minimum sample size needed for *t*-test was N = 65. Thus, the obtained sample size of N = 90 is sufficient for this study.

3.1.2. Materials and stimuli

For Experiment 2, two new soundtracks of 30-second length (C1 and C2) were composed based on the musical features obtained in Experiment 1 (F3 and F1). These soundtracks embodied the brand values, representing nature, cool, cold, humid, light, calm, outdoor exploration, and well-being, in an elegant and original manner. These stimuli are not publicly accessible due to copyright restrictions.

3.1.3. Measures

During Experiment 2, one main variable assessed was the reaction time for each primer term on each freshness and brand archetype attribute (section 3.1.4). The reaction time was used to compute the association index, indicating the latency at which participants pressed the response key to measure the strength of association between the sound and an attribute (Greenwald et al., 1998; Teichert et al., 2019). If participants implicitly agreed with the combination, they pressed “YES” faster; if they disagreed, they pressed “NO” faster. Each primer term was presented twice, once for “YES” and once for “NO,” as this is not a choice test.

3.1.4. Experimental design and procedure

Experiment 2 followed a within-participants approach to compare the implicit associations evoked by each sound. Participants experienced three auditory stimuli: the two soundtracks described in section 3.1.2, and white noise as a control.

This experiment was administered online and lasted approximately 15 min. Participants signed up voluntarily through the recruiting agency (Offerwise). First, they were instructed to use headphones and underwent an auditory test to ensure proper usage.

Next, participants read the following instructions: *“Welcome to the game of identifying the words. The objective is to try to finish as fast as possible. How fast can you answer? Your task is: When you see “YES,” press M. When you see “NO,” press C. Do it as fast as you can! If you press the wrong key, your computer will emit a beep.”*.

The keys for “YES” and “NO” were randomly assigned between ’M’ and ’C.’ Prior to the main test, participants received training in the IAT by pressing the keys according to the word on the screen until they correctly pressed each key four times. “YES” and “NO” appeared randomly twice each during the training.

During the main task, participants listened to the three sounds (white noise, C1, and C2) randomly. While listening to one sound, they were instructed to focus on it. After listening to the entire sound (30 s), participants completed the main IAT. The text “this sound is” alongside a term associated with freshness (i.e., fresh, cold, humid, smooth, blue, or light) or the brand archetype (i.e., watery, energetic, natural, or resembles cleanliness; see section 1.2 for more details on these attributes) appeared briefly (500 ms), followed by a “YES” or “NO” instruction indicating which key to press (C or M; see Fig. 4). This process was repeated until each term appeared twice (one for “YES” and one for “NO”), resulting in 20 combinations in random order. All the process was repeated with the two remaining sounds. The experiment was conducted in Brazilian Portuguese.

3.1.5. Data analysis

To assess statistical significance, one-sample t-tests were used to compare the association index for each attribute against a random score of 50 % (representing the probability of correctly detecting “YES” or “NO”). The t-tests analyzed the reaction times of the sample to determine if they significantly differed from random, either positively or negatively.

3.2. Results

In Experiment 2, the soundtrack C2 was positively associated with the average of attribute associations in freshness and brand archetype dimensions (average of the attributes included in each dimension). C1 was positively associated with the average in the freshness dimension but negatively associated with the brand archetype dimension. White noise did not differ from the random score on either dimension average, indicating a neutral association.

Concerning individual freshness attributes (see Fig. 5), C2 was positively associated with light, cold, and smooth attributes (with a rather large positivity for light), but negatively associated with humid and blue ($p < 0.05$). C1 was positively correlated to fresh, cold, blue, and light attributes, but negatively correlated to humid and smooth ($p < 0.05$). White noise was positively linked to cold, smooth, and light attributes, but negatively linked to humid and blue ($p < 0.05$).

Regarding the brand archetype dimension (Fig. 5), C2 was positively associated with natural and cleanliness attributes, but negatively associated with watery ($p < 0.05$). C1 was negatively correlated to all four attributes ($p < 0.05$). White noise was positively linked to energetic and natural attributes, but negatively linked to watery and cleanliness ($p < 0.05$).

4. Experiment 3

Experiment 3 aimed to validate the multisensory interactions between scent and hearing by evaluating the ability of the commercial soundtracks C1 and C2, from Experiment 2, to enhance the perception of freshness of two selected commercial fragrances. Male participants completed an IAT in the laboratory across freshness and brand archetype dimensions while being stimulated by the scents. The analysis examined

Fig. 4. Screenshots of the IAT from Experiment 2. Left: term associated with freshness or brand ("this sound is blue"). Right: "YES" or "NO" to indicate the participant to press C or M key ("YES"). The keys assigned to "YES" and "NO" were randomly chosen between C and M at the beginning of the test. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

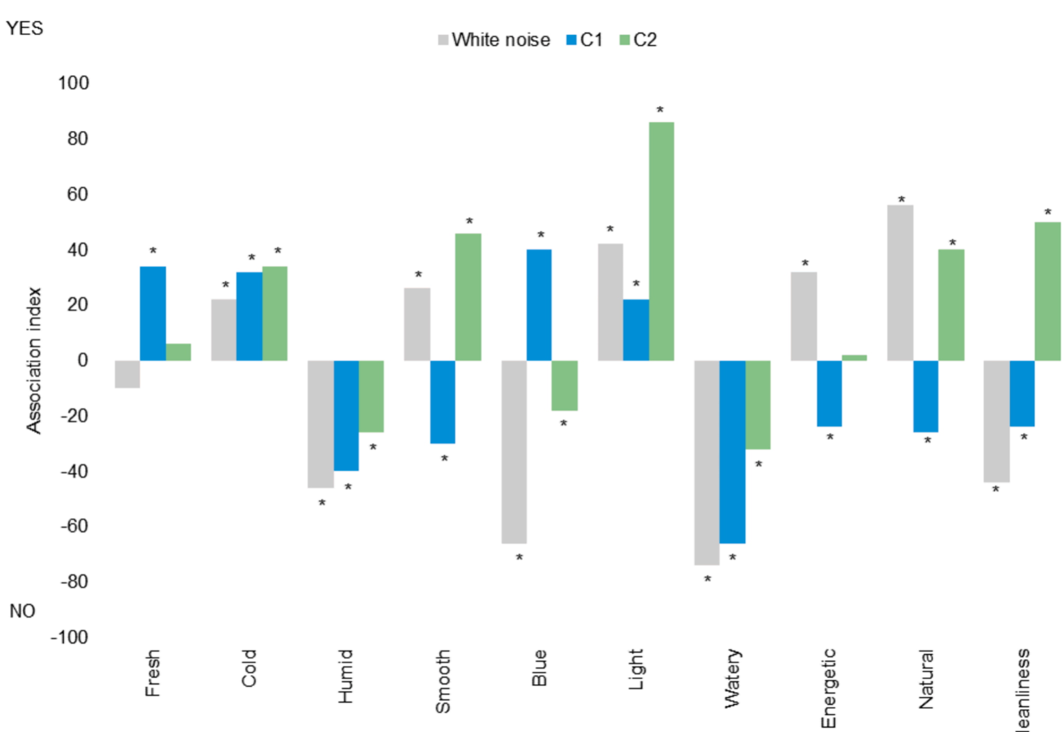

Fig. 5. Association indexes regarding the reaction times around the terms of the freshness and brand archetype dimensions after exposure to sounds used in Experiment 2. A positive association index means that the corresponding sound is positively associated with the attribute in the x-axis. On the contrary, a negative association index means that the corresponding sound is negatively associated with the attribute. Asterisk means that the corresponding association index differs significantly from a random score (i.e., the corresponding sound is significantly associated or not with that attribute; p-value < 0.05).

implicit scent perception without sound and with each soundtrack. Note that the IAT has been successfully used to evaluate olfactory perceptions (e.g., Demattè et al., 2006; Demattè et al., 2007; Lemercier-Talbot et al., 2019; Stafford et al., 2019).

4.1. Methodology

4.1.1. Participants

Two hundred participants (male, aged 29–49 years, M = 37 years, SD = 6) joined Experiment 3 in a laboratory in São Paulo, Brazil. All participants received $120 Brazilian reals. They were regular perfume users (80 % of the target brand).

An a priori power analysis using G*Power 3.1.9.7 (Faul et al., 2007) determined the minimum sample size. With an effect size of 0.5 (medium size according to Cohen, 1988), $\alpha = 0.05$, and power = 0.99, the minimum sample size needed is for two-sample *t*-test N = 176. Thus, the obtained sample size of N = 200 is adequate for this study.

4.1.2. Materials and stimuli

Experiment 3 used the commercial soundtracks from Experiment 2, delivered *via* headphones Philips HiFi stereo SHP2700. The olfactory stimuli consisted of two brand's fragrances (Fragrance1 and Fragrance2). According to fragrance classification, Fragrance1 and Fragrance2 both belong to the *fougère* family but differ in subgroups and notes. Fragrance1 represents the "water" subgroup with a modern freshness. It features cold spicy cardamom and nutmeg top notes, followed by mint and rosemary. The heart maintains this freshness with watery notes and cashmeran wood, while the base offers warmth with cedarwood, musk, and amber. In contrast, Fragrance2 offers a traditional post-bathing freshness. Its top notes include citrus and metallic touches, with musk in the base notes. Lavender and neroli provide aromatic strength in the heart notes, and dry woods add elegance to the base. Gas chromatography coupled with mass spectrometry (CG-MS) profiles showed distinct characteristics: Fragrance1 with fresh citrus, floral sweet tones, and woody notes, whereas Fragrance2 with floral watery notes and a hint of camphoraceous. Both fragrances have a 9–15 % fragrance concentration, suitable for the Brazilian market. Trade-grade scent strips were promptly given to participants after minimal immersion in fragrance containers.

4.1.3. Measures

The dependent variable in Experiment 3 was the reaction time for each primer term on each freshness and brand archetype attribute (section 3.1.4), serving as the basis for calculating the association index. In this case, the fragrance served as the target concept, potentially associated with each attribute dimension.

4.1.4. Experimental design

Experiment 3 used a mixed-model design to examine the effect of the same soundtrack on perceptions of different fragrances. As shown in Fig. 6, participants were initially exposed to one scent (50 % to Fragrance1 and 50 % to Fragrance2 randomly), followed by random allocation to one of two groups. One group experienced the C1 soundtrack paired with one scent and then with the other scent (in random order). The other group experienced C2 paired with one scent and then with the other scent (in random order). Thus, the first stage involved two between-participants conditions based on the scent, while the second stage had between-participants conditions for the soundtracks and within-participants conditions for the scents.

4.1.5. Procedure

Participants provided free and informed consent and spent an average of 50 min completing the test. They underwent auditory and olfactory acuity tests. The latter was performed with gauzes soaked with water and ethyl alcohol solution concentrations of 10 %, 25 %, 50 %, 70 %, and 96 %, and only those who managed to distinguish at least from the 50 % concentration were considered in the test (Fenólio et al., 2020). Those who passed the tests proceeded with the survey on a computer using headphones. The instructions and IAT training phase followed the procedure of Experiment 2 (section 3.1.4).

During the main task, a scent strip trade grade was delivered to each participant with the corresponding scent while listening to no sound (Fig. 6). After smelling the scent (Fig. 7) for 30 s, participants completed an IAT. The text "this fragrance is" alongside a freshness- and brand-related term, described in Experiment 2 (section 3.1.4), appeared for 500 ms. Participants pressed the C or M key according to the "YES" or "NO" text displayed. This process repeated for each term twice (one for "YES" and one for "NO") in random order, resulting in 20 combinations. All the process was repeated with one scent paired with the corresponding soundtrack and then the other scent with the same soundtrack (Fig. 6). Participants smelled water between scent stimuli to minimize olfactory saturation. The experiment was conducted in Brazilian Portuguese.

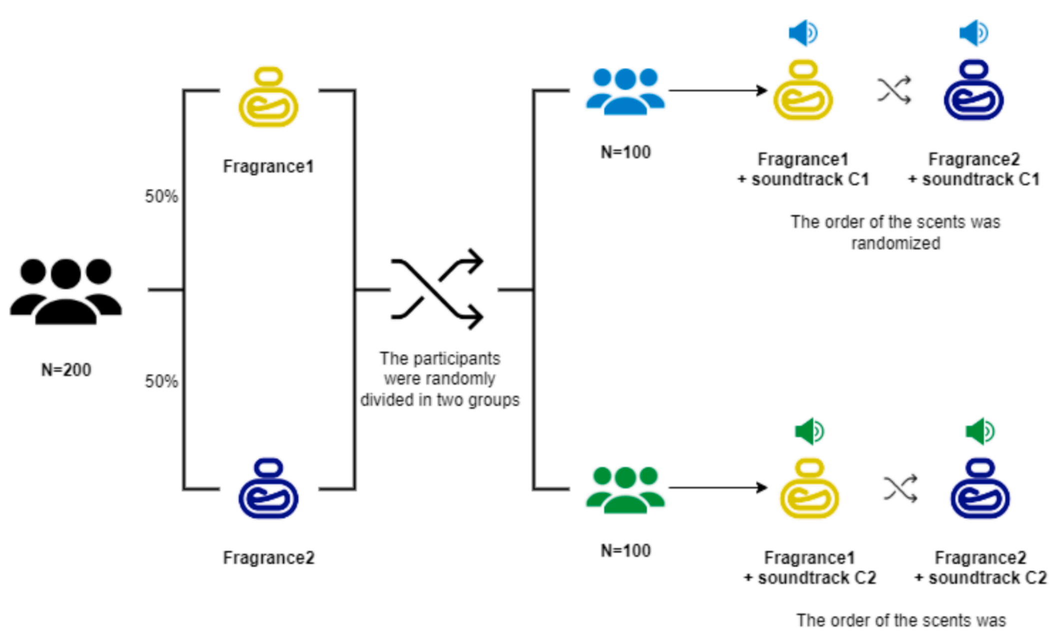

Fig. 6. Illustration of the experimental design in Experiment 3, including the two types of stimuli (scent and sound).

Fig. 7. A participant smelling a scent stimulus during Experiment 3.

4.1.6. Data analysis

The statistical analysis in Experiment 3 involved the tests conducted in Experiment 2. Additional t-tests compared the effects of scents in their control state (without concurrent sound) to the effects of scents accompanied by each soundtrack.

4.2. Results

In the absence of soundtracks, none of the fragrances differed significantly from the random score on the average of attribute associations of freshness dimension (average of the attributes included in this dimension), indicating a neutral association. Conversely, on the average

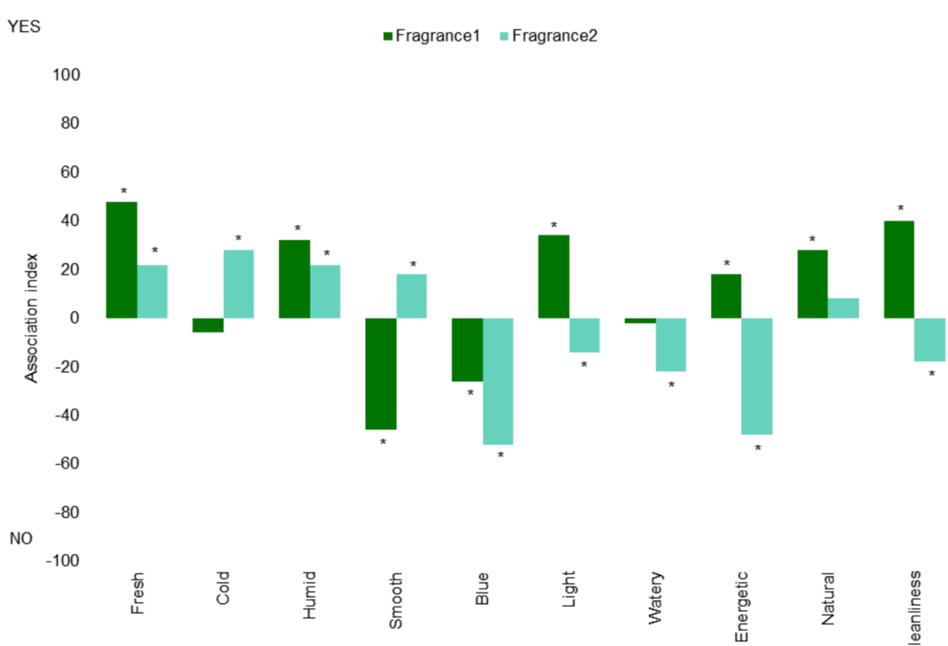

Fig. 8. Association indexes regarding the reaction times around the terms of the freshness and brand archetype dimensions after exposure to scents in Experiment 3. A positive association index means that the corresponding sound is positively associated with the attribute in the x-axis. On the contrary, a negative association index means that the corresponding sound is negatively associated with the attribute in the x-axis. Asterisk means that the corresponding association index differs significantly from a random score (i.e., the corresponding soundtrack is significantly associated or not with that attribute, regarding the sign of the index; p-value < 0.05).

of attribute associations of brand archetype dimension, Fragrance1 showed a positive association, whereas Fragrance2 showed a negative association. Among the freshness terms (see Fig. 8), the Fragrance1 scent was positively associated with fresh, humid, and light, but negatively associated with smooth and blue ($p < 0.05$). Fragrance2, on the other hand, was positively correlated to fresh, cold, humid, and smooth attributes, but negatively correlated to blue and light ($p < 0.05$). Among the brand archetype terms (Fig. 8), Fragrance1 was positively linked to energetic, natural, and cleanliness attributes, whereas Fragrance2 was negatively linked to watery, energetic, and cleanliness ($p < 0.05$).

When it comes to the effects of soundtracks on Fragrance1 perception, C2 turned the neutral association with the average in freshness dimension into a positive one, while maintaining the positive association with the brand archetype dimension. Conversely, the soundtrack C1 turned the neutral association of Fragrance1 with the average in freshness dimension into a negative one and the positive association with the brand archetype dimension into a neutral one.

In terms of individual freshness attributes in the Fragrance1 perception (see Fig. 9), the inclusion of C2 significantly increased the implicit perception of cold and blue attributes, resulting in positive associations ($p < 0.05$). However, it significantly decreased the perception of the fresh attribute while maintaining a positive association ($p < 0.05$). The inclusion of C1 increased the perception of the smooth attribute while maintaining a negative association ($p < 0.05$). Additionally, C1 led to decreased perceptions of fresh and light attributes (maintaining a positive association), as well as humid (resulting in a neutral association) and cold (resulting in a negative association; $p < 0.05$). Within the brand archetype dimension (Fig. 9), the inclusion of C2 significantly increased the implicit perception of the energetic attribute, maintaining a positive association ($p < 0.05$). However, it significantly decreased the perception of cleanliness and watery attributes, resulting in a neutral and negative association, respectively ($p < 0.05$). The inclusion of C1 decreased perceptions of watery and natural attributes, leading to a neutral and a negative association, respectively ($p < 0.05$).

Concerning the effects of soundtracks on Fragrance2 perception, C1 turned its neutral association with the average in freshness dimension into a negative one and the negative association with the average in brand archetype dimension into a neutral one. However, including the soundtrack C2 did not alter the direction of the associations on either dimension.

Among the freshness attribute associations in the Fragrance2 perception (see Fig. 10), the inclusion of C2 significantly increased the implicit perception of blue and light attributes, turning into positive associations ($p < 0.05$). However, it decreased the perception of fresh and cold, turning into negative associations ($p < 0.05$). The inclusion of C1 increased the perception of blue (maintaining a negative association) and light (turning into a positive association), while decreasing the perceptions of cold, humid, and smooth, turning into negative associations ($p < 0.05$). Within the brand archetype dimension (Fig. 10), the inclusion of C2 significantly increased the implicit perception of the energetic attribute, turning into a positive association ($p < 0.05$). However, it significantly decreased the perception of cleanliness and watery, maintaining negative associations, while increasing the perception of natural, shifting from a neutral to a negative association ($p < 0.05$). The inclusion of C1 increased the perception of energetic and cleanliness, removing negative associations ($p < 0.05$). It also decreased the perceptions of natural and watery, maintaining their neutral and negative associations, respectively ($p < 0.05$).

5. Discussion

This study investigated the impact of brand-representative sounds on the perception of freshness in fragrances. Inspired by the possibility of enabling cosmetics retailing to prompt fresh sensations more effectively through sounds, this study used two fragrances from a commercial perfume brand to test the auditory modulation around the olfactory experience. Self-reported and implicit responses (*via* IAT) around attributes related to freshness were examined for auditory and olfactory

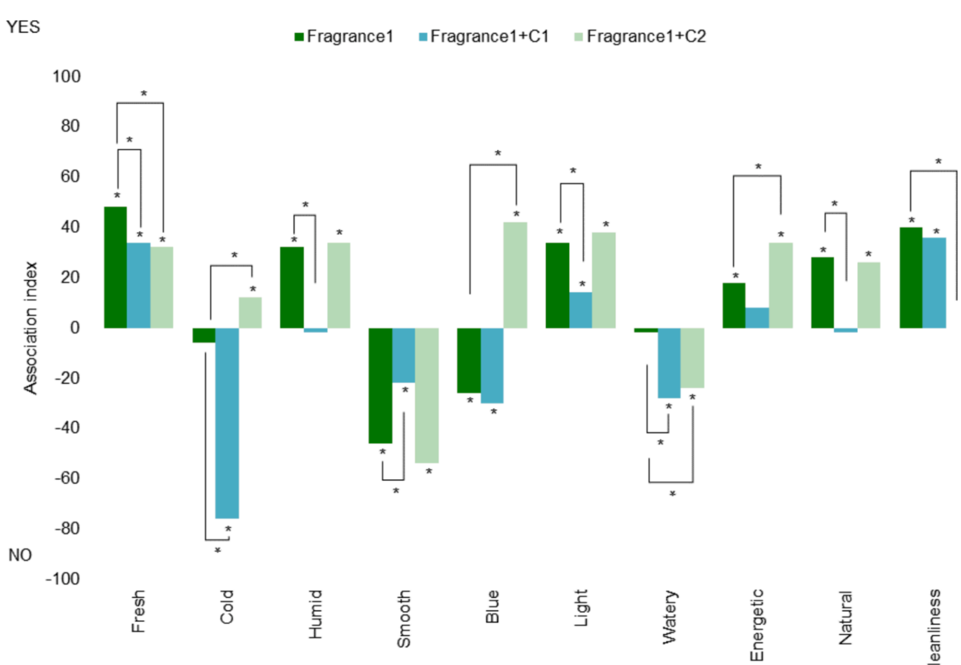

Fig. 9. Association indexes regarding the reaction times around the terms of the freshness and brand archetype dimensions after exposure to Fragrance1 scent with no sound and along with C1 and C2 soundtracks in Experiment 3. A positive association index means that the corresponding sound is positively associated with the attribute in the x-axis. On the contrary, a negative association index means that the corresponding sound is negatively associated with the attribute in the x-axis. Asterisk above the column means that the corresponding association index differs significantly from a random score (i.e., the corresponding soundtrack is significantly associated or not with that attribute, regarding the sign of the index; p-value < 0.05). An asterisk relating two columns means that the two association indexes differ significantly from each other (p-value < 0.05).

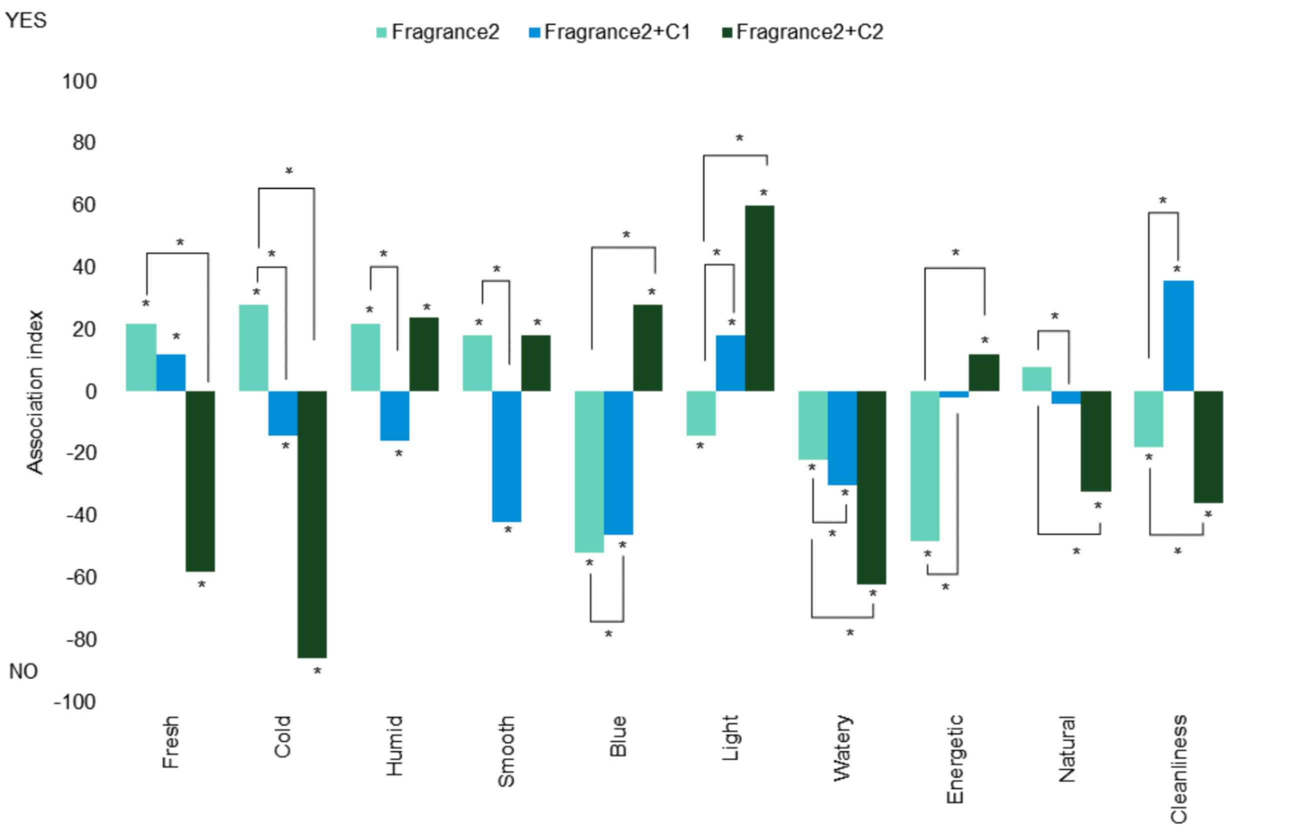

Fig. 10. Association indexes regarding the reaction times around the terms of the freshness and brand archetype dimensions after exposure to Fragrance 2 scent alone and with C1 and C2 sounds in Experiment 3. A positive association index means that the corresponding sound is positively associated with the attribute in the x-axis. On the contrary, a negative association index means that the corresponding sound is negatively associated with the attribute in the x-axis. Asterisk above the column means that the corresponding association index differs significantly from a random score (i.e., the corresponding soundtrack is significantly associated or not with that attribute, regarding the sign of the index; p-value < 0.05). An asterisk relating two columns means that the two association indexes differ significantly from each other (p-value < 0.05).

experiences.

The study introduced innovative methodologies to enhance understanding in this field. Firstly, baseline auditory stimuli were carefully selected based on finding parameters associated with freshness like temperature, humidity, emotional valence, olfactory notes, and colors. These parameters were then paired with specific musical characteristics, including frequency ranges, timbres of musical instruments, musical genre, musical tempo, and nature sounds, based on the existing literature on crossmodal associations. Secondly, implicit association measures were mostly aimed at detecting underlying automatic associations, reflecting the system's responses, not only to conscious but also to unconscious processes (Greenwald et al., 1998). IAT is suitable for detecting automatic associations involving pure scents (i.e., with monomolecular compounds) as well as with commercial fragrances (i.e., complex mix of scents), which are complex volatile mixtures (Lemercier-Talbot et al., 2019). In this study, such a method revealed whether the soundtrack or fragrance was positively or negatively associated with primer terms without the need for control or contrast conditions.

The findings demonstrated that music modulated the perception of individual freshness attributes, which were tested separately due to the lack of consensus in the freshness olfactory description (Zarzo, 2012). Experiment 1′s results showed the perceived correspondence between sounds with specific features and olfactory freshness attributes across a large sample, with two sounds standing out and being more in line with brand desires. After composing new commercial soundtracks based on the latter, Experiment 2 provided evidence for the replications of the results in implicit associations. Finally, Experiment 3 showed that the Fragrance1 scent became perceived as cold and blue and Fragrance2 as light and blue when accompanied by one of the composed soundtracks. These outcomes uncovered the crucial role of such musical features in enhancing the freshness experience of commercial fragrances.

Specifically, the C2 soundtrack consistently triggered the initial requirements of the brand. It successfully conveyed relevant and specific freshness attributes (i.e., light and humid) for the fragrances (Fig. 11). Moreover, the blue color became subconsciously present in the consumers' minds while experiencing fragrances accompanied by this soundtrack, although both fragrances were negatively associated with blue when assessed without any sound. Therefore, the implicit association of the fragrances with blue was constructed by music, more specifically by the C2 soundtrack. While there is a long history of sensory translation between audition and vision (see Spence & Di Stefano, 2023, for a recent review), this novel finding highlights the powerful role of

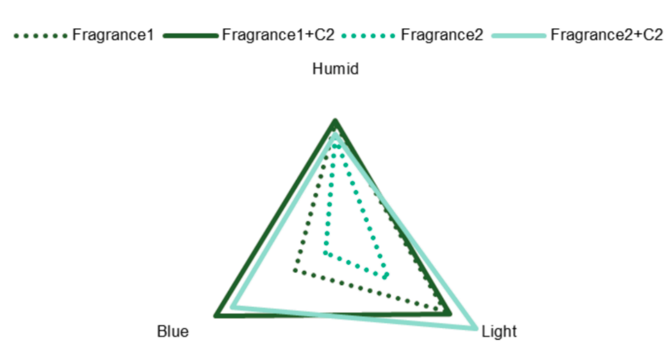

Fig. 11. Comparison of the association indexes between the fragrances without sound and accompanied by the C2 soundtrack and the top three attributes of the fresh experience in Experiment 3.

music while uncovering crossmodal correspondences with specific colors in olfactory experiences, as this correspondence has not been as obvious.

The existing multisensory research has highlighted the challenges of experimentally assessing the impact of music on olfactory note perception. Previous studies have primarily focused on the emotional/hedonic aspects of scent perception influenced by music (Spence, 2021). Additionally, reaching statistical significance when evaluating the influence of music on aroma perception has been acknowledged as a challenge (Velasco et al., 2014). The present study overcame such difficulties while assessing the modulatory effect of sound on olfactory perception, revealing important findings concerning the olfactory freshness experience.

Prior literature indicates that the IAT might capture similarities between stimuli through both semantic memory and perceptual salience (De Houwer et al., 2005). Semantic similarity is rooted in shared meaning, while perceptual salience involves features influencing similarity, irrespective of semantics. Our results suggest a dual explanation; some sound or fragrance features evoke semantic congruency (as in natural and watery), while others highlight perceptual salience (as in light and humid). This dual lens clarifies the factors contributing to the formation of certain associations.

In both Experiments 1 and 2, specific attributes, such as light and cold, consistently maintained positive ratings concerning the fresh sounds, whereas other attributes, including humidity, showed variations, confirming that explicit and implicit measures are not always positively correlated (Gawronski & Hahn, 2018). Actually, the mechanisms by which individuals associate information between two sensory modalities may be diverse when talking about conscious and automatic processes (Fazio & Olson, 2003; Gawronski & Houwer, 2014; Haase & Wiedmann, 2020; Lemercier-Talbot et al., 2019). Therefore, applying only conscious measures may represent results with the bias of self-awareness, excluding a part of the truth (Dijksterhuis et al., 2005; Haase & Wiedmann, 2020).

This study also acknowledges the potential limitations of sounds when attempting to directly modulate actual olfactory experiences. The smell is a “chemical sense” (Alfonso-Prieto, 2021) that seems to encompass about 400 olfactory receptors through which humans may be able to detect around 5000 discriminable olfactory stimuli considering moderately conservative statistical criteria (Gerkin & Castro, 2015), making olfactory coding complex (Korsching, 2013). Hence, these transfer effects are not generic when attempting to modulate an actual olfactory experience. For example, in Experiment 2, the C1 soundtrack was negatively associated with the resemblance of cleanliness, while C2 was positively associated with it. However, in Experiment 3, C1 increased the cleanliness perception of Fragrance2, whereas C2 decreased this perception. Furthermore, Fragrance1 showed a negative association with smoothness, and the inclusion of C1 increased this association but kept it negative. In contrast, the same soundtrack decreased the positive association for Fragrance2, shifting it to negative. On top of that, Fragrance1 showed a positive association with lightness, and the inclusion of C1 decreased this association but still kept it positive, whereas the same soundtrack increased the negative association for Fragrance2, shifting it to positive. Thus, C1 and C2 modulate the perception of each scent differently, with C2 producing positive effects on various attributes related to fresh experiences for both scents. These results suggest that the initial point of the olfactory experience seems to be influential in multisensory integration.

Sonic seasoning (Spence et al., 2021) in cosmetics may help innovate when it comes to conveying complex commercial fragrance experiences more effectively. Visual cues like colors and shapes have already been reported as an opportunity to address this challenge in olfactory perception (Spence, 2020), but the sound has been scarcely studied, with existing studies mostly framed in the context of food (Rodriguez et al., 2023). The cosmetic industry can certainly benefit from expanding the scope of multisensory experiences around olfaction, further innovating through science.

5.1. Implications for practitioners

Most brands overlook the potential of sound and music as impactful attributes for brand awareness and preference (Arora & Kumar, 2018; Reinoso-Carvalho et al., 2019). Nonetheless, in the perfumery industry, some manufacturers have started to innovatively incorporate sounds into their promotion strategies. Recently, the fragrance Spicebomb Infrared by Viktor&Rolf in collaboration with L’Oréal Luxe Division featured an original soundtrack created by Ircam Amplify, selecting sounds that conveyed the perfume’s hot and spicy notes (Edelson, 2021; L’Oréal Group, 2021). Another fragrance brand, Room 1015, known for its nostalgic fragrances inspired by the era of rock ’n’ roll, crafts a unique approach to perfumery by creating a soundtrack for each fragrance, combining music and scent (ROOM 1015, n.d.). Similarly, L’Orchestre Parfum offers a range of fragrances inspired by music, providing a dedicated soundtrack for each perfume (L’Orchestre Parfum, 2021).

Note that the aforementioned marketing approaches rely mostly on sound lexicons, which combine words and sound samples, and on the designer’s expertise, thereby disregarding a more consumer-centric and experimental approach. Our study goes beyond by framing music as an added value to a consumer and brand’s multisensory perception, particularly for perfume brands. Therefore, this study provides insights when it comes to connecting with consumers in innovative ways, while also disrupting the existing needs of the beauty industry. For example, music with woodwind and piano instrumentation, low tempo, and legato articulation can evoke the desired sensations of freshness. By incorporating commercial validity, these novel results can motivate the cosmetic industry seeking to augment freshness to consider soundscape during phygital customer product experience while supporting brand positioning.

To convey sensations other than freshness, future studies could apply the protocol outlined in section 1.1. This process can facilitate the conveyance of specific sensations, but also explore different competitive positioning strategies that leverage complex sonic identities, contributing to increased market value and consumer enchantment (Arora & Kumar, 2018; Reinoso-Carvalho et al., 2019). Therefore, this study contributes to establishing methodological foundations for aligning brand-aligned soundtracks with specific attributes in product experiences.

The use of the IAT in marketing research enables the collection of information about how brand sensory elements are associated, without requiring explicit awareness of crossmodal correspondences. Prior research has found that the consumer’s unconscious mind, constantly gathering information, influences decision-making (Haase & Wiedmann, 2020; Parise & Spence, 2012). The IAT has successfully unveiled both implicit and explicit semantic associations between brand packaging and brand attributes (e.g., Listerine with “powerful”; Parise & Spence, 2012). On top of that, by revealing implicit and explicit associations of music and fragrance with brand attributes, our findings provide insights for creating tailored and effective multisensory experiences around perfume brands.

With the growing popularity of online shopping for cosmetics (Liu et al., 2013), communicating fragrance sensations through auditory and visual cues becomes critical to building product expectations due to the limitations of the current digital experiences. Since emotional trust, valence, and hedonic values influence cosmetics online purchase intention (H. S. Lee et al., 2015; Suparno, 2020; Sutanto & Aprianingsih, 2016), brands may feel encouraged to use customized soundtracks, creating immersive experiences engaging consumer emotions. On the other hand, physical retail might focus on providing meaningful in-store experiences that captivate consumers through multisensory olfactory encounters. Adding value to the experience of consumers through different channels can break down physical barriers by connecting the senses of smell and hearing.

5.2. Limitations

While the IAT seems a powerful tool for enabling multisensory integrations without overtly asking people about their product experience, it has some methodological limitations. Recent research has highlighted potentially problematic aspects, including its inability to isolate implicit associations (since it also measures explicit ones), its susceptibility to extraneous influences, and its limited behavior predictive validity (Meißner et al., 2019; Röhner & Iliescu, 2023; Schimmack, 2021). These issues could lead to miscalculation of the strength of implicit associations. Therefore, exploring alternative scoring algorithms, considering the degree of cognitive control, and discriminating dependent variables of IAT are needed to understand the underlying mechanisms of such measures.

Our results indicate that white noise might not be an appropriate control for olfactory sensations. In Experiment 2, white noise showed positive associations with cold, smooth, light, energetic, and natural attributes, while having negative associations with humid, blue, watery, and cleanliness. In a previous study, white noise was consciously rated as unpleasant, and some aromas presented after this noise were rated as less pleasant, sweet, and drier compared to musical conditions (Velasco et al., 2014). Future research should explore alternative control auditory conditions for olfactory experiences.

The fragrances used in this study were complex olfactory mixtures with commercial orientation and, thus, formed by the harmonious combination of numerous ingredients of different chemical characteristics, volatility, and sources. Moreover, some of the definitions that we used to frame these fragrances may be subject to cultural differences (i. e., Blank & Mattes, 1990). Using these fragrances certainly provides the study with somewhat realistic/commercial validity. However, there is an opportunity to deepen the understanding of the multisensory effects assessed in this study by considering the analysis of isolated ingredients *versus* the complete fragrance. The latter would allow us to further understand the impact of each olfactory ingredient in this crossmodal relationship.

6. Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the work. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

7. Declaration of Ethics

All participants agreed with informed consent prior to joining the study, which was approved by the ethics committee of Escola Superior de Propaganda e Marketing (ESPM) under Act 58723022.7.0000.9127 of 2022.

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CRediT authorship contribution statement

Brayan Rodríguez: Conceptualization, Methodology, Investigation, Data Curation, Writing – original draft, Visualization. **Monique Alves Frazon Cantu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Luis H. Reyes:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Vanessa Jaqueline De Almeida Ribas Pereira:** Resources, Investigation, Conceptualization. **Larissa Carmona Zonta Santos:** Writing – review & editing, Resources, Project administration. **Felipe Reinoso-Carvalho:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Felipe Reinoso-Carvalho reports financial support was provided by Boticario Group. Monique Alves Frazon Cantu, Larissa Carmona Zonta, Vanessa Jaqueline de Almeida Ribas Pereira reports a relationship with Boticario Group that includes: employment.

Data availability

The data that has been used is confidential.

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Appendix

Table A1
Pairwise comparisons between auditory conditions on each sensation of Experiment 1.

Dependent Variable	Stimuli		Mean Difference	p-value
Cold	F5	F6	-0,02	1,00
		F1	-0,40	0,92
		F2	0,39	1,00
		F3	-0,26	1,00
		F4	-0,34	1,00
		NF1	1,47	0,00
		NF2	1,19	0,00
	F6	F5	0,02	1,00

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F1	-0,38	1,00
		F2	0,41	0,75
		F3	-0,24	1,00
		F4	-0,32	1,00
		NF1	1,49	0,00
		NF2	1,21	0,00
	F1	F5	0,40	0,92
		F6	0,38	1,00
		F2	0,79	0,00
		F3	0,14	1,00
		F4	0,06	1,00
		NF1	1,87	0,00
		NF2	1,59	0,00
	F2	F5	-0,39	1,00
		F6	-0,41	0,75
		F1	-0,79	0,00
		F3	-0,66	0,01
		F4	-0,73	0,00
		NF1	1,08	0,00
		NF2	0,80	0,00
	F3	F5	0,26	1,00
		F6	0,24	1,00
		F1	-0,14	1,00
		F2	0,66	0,01
		F4	-0,07	1,00
		NF1	1,74	0,00
		NF2	1,46	0,00
	F4	F5	0,34	1,00
		F6	0,32	1,00
		F1	-0,06	1,00
		F2	0,73	0,00
		F3	0,07	1,00
		NF1	1,81	0,00
		NF2	1,53	0,00
	NF1	F5	-1,47	0,00
		F6	-1,49	0,00
		F1	-1,87	0,00
		F2	-1,08	0,00
		F3	-1,74	0,00
		F4	-1,81	0,00
		NF2	-0,28	1,00
	NF2	F5	-1,19	0,00
		F6	-1,21	0,00
		F1	-1,59	0,00
		F2	-0,80	0,00
		F3	-1,46	0,00
		F4	-1,53	0,00
		NF1	0,28	1,00
Humid	F5	F6	-0,07	1,00
		F1	-1,40	0,00
		F2	-0,10	1,00
		F3	-0,85	0,00
		F4	-0,88	0,00
		NF1	-0,05	1,00
		NF2	0,14	1,00
	F6	F5	0,07	1,00
		F1	-1,33	0,00
		F2	-0,03	1,00
		F3	-0,78	0,00
		F4	-0,81	0,00
		NF1	0,03	1,00
		NF2	0,21	1,00
	F1	F5	1,40	0,00
		F6	1,33	0,00
		F2	1,30	0,00
		F3	0,55	0,08
		F4	0,52	0,16
		NF1	1,36	0,00
		NF2	1,54	0,00
	F2	F5	0,10	1,00
		F6	0,03	1,00
		F1	-1,30	0,00
		F3	-0,75	0,00
		F4	-0,78	0,00

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		NF1	0,05	1,00
		NF2	0,24	1,00
	F3	F5	0,85	0,00
		F6	0,78	0,00
		F1	-0,55	0,08
		F2	0,75	0,00
		F4	-0,03	1,00
		NF1	0,80	0,00
		NF2	0,99	0,00
	F4	F5	0,88	0,00
		F6	0,81	0,00
		F1	-0,52	0,16
		F2	0,78	0,00
		F3	0,03	1,00
		NF1	0,84	0,00
		NF2	1,02	0,00
	NF1	F5	0,05	1,00
		F6	-0,03	1,00
		F1	-1,36	0,00
		F2	-0,05	1,00
		F3	-0,80	0,00
		F4	-0,84	0,00
		NF2	0,19	1,00
	NF2	F5	-0,14	1,00
		F6	-0,21	1,00
		F1	-1,54	0,00
		F2	-0,24	1,00
		F3	-0,99	0,00
		F4	-1,02	0,00
		NF1	-0,19	1,00
Outdoor	F5	F6	-0,56	0,36
		F1	-1,80	0,00
		F2	-1,13	0,00
		F3	-0,79	0,01
		F4	-0,83	0,01
		NF1	0,58	0,30
		NF2	1,04	0,00
	F6	F5	0,56	0,36
		F1	-1,24	0,00
		F2	-0,57	0,29
		F3	-0,23	1,00
		F4	-0,26	1,00
		NF1	1,14	0,00
		NF2	1,60	0,00
	F1	F5	1,80	0,00
		F6	1,24	0,00
		F2	0,67	0,06
		F3	1,02	0,00
		F4	0,98	0,00
		NF1	2,38	0,00
		NF2	2,84	0,00
	F2	F5	1,13	0,00
		F6	0,57	0,29
		F1	-0,67	0,06
		F3	0,34	1,00
		F4	0,30	1,00
		NF1	1,71	0,00
		NF2	2,17	0,00
	F3	F5	0,79	0,01
		F6	0,23	1,00
		F1	-1,02	0,00
		F2	-0,34	1,00
		F4	-0,04	1,00
		NF1	1,37	0,00
		NF2	1,82	0,00
	F4	F5	0,83	0,01
		F6	0,26	1,00
		F1	-0,98	0,00
		F2	-0,30	1,00
		F3	0,04	1,00
		NF1	1,41	0,00
		NF2	1,86	0,00
	NF1	F5	-0,58	0,30
		F6	-1,14	0,00

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F1	-2,38	0,00
		F2	-1,71	0,00
		F3	-1,37	0,00
		F4	-1,41	0,00
		NF2	0,46	1,00
	NF2	F5	-1,04	0,00
		F6	-1,60	0,00
		F1	-2,84	0,00
		F2	-2,17	0,00
		F3	-1,82	0,00
		F4	-1,86	0,00
		NF1	-0,46	1,00
Pleasant	F5	F6	0,04	1,00
		F1	-0,53	0,23
		F2	-1,11	0,00
		F3	-0,45	0,74
		F4	-0,40	1,00
		NF1	-0,27	1,00
		NF2	0,29	1,00
	F6	F5	-0,04	1,00
		F1	-0,57	0,11
		F2	-1,15	0,00
		F3	-0,49	0,40
		F4	-0,44	0,81
		NF1	-0,31	1,00
		NF2	0,25	1,00
	F1	F5	0,53	0,23
		F6	0,57	0,11
		F2	-0,58	0,09
		F3	0,08	1,00
		F4	0,13	1,00
		NF1	0,26	1,00
		NF2	0,82	0,00
	F2	F5	1,11	0,00
		F6	1,15	0,00
		F1	0,58	0,09
		F3	0,66	0,03
		F4	0,71	0,01
		NF1	0,84	0,00
		NF2	1,40	0,00
	F3	F5	0,45	0,74
		F6	0,49	0,40
		F1	-0,08	1,00
		F2	-0,66	0,03
		F4	0,05	1,00
		NF1	0,18	1,00
		NF2	0,74	0,01
	F4	F5	0,40	1,00
		F6	0,44	0,81
		F1	-0,13	1,00
		F2	-0,71	0,01
		F3	-0,05	1,00
		NF1	0,13	1,00
		NF2	0,69	0,02
	NF1	F5	0,27	1,00
		F6	0,31	1,00
		F1	-0,26	1,00
		F2	-0,84	0,00
		F3	-0,18	1,00
		F4	-0,13	1,00
		NF2	0,56	0,14
	NF2	F5	-0,29	1,00
		F6	-0,25	1,00
		F1	-0,82	0,00
		F2	-1,40	0,00
		F3	-0,74	0,01
		F4	-0,69	0,02
		NF1	-0,56	0,14
Comfortable	F5	F6	0,48	0,48
		F1	-0,36	1,00
		F2	-0,76	0,00
		F3	-0,27	1,00
		F4	-0,09	1,00

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		NF1	0,17	1,00
		NF2	1,02	0,00
	F6	F5	-0,48	0,48
		F1	-0,84	0,00
		F2	-1,24	0,00
		F3	-0,75	0,00
		F4	-0,57	0,11
		NF1	-0,30	1,00
		NF2	0,54	0,15
	F1	F5	0,36	1,00
		F6	0,84	0,00
		F2	-0,40	1,00
		F3	0,09	1,00
		F4	0,27	1,00
		NF1	0,53	0,18
		NF2	1,38	0,00
	F2	F5	0,76	0,00
		F6	1,24	0,00
		F1	0,40	1,00
		F3	0,49	0,34
		F4	0,67	0,02
		NF1	0,94	0,00
		NF2	1,78	0,00
	F3	F5	0,27	1,00
		F6	0,75	0,00
		F1	-0,09	1,00
		F2	-0,49	0,34
		F4	0,18	1,00
		NF1	0,44	0,66
		NF2	1,29	0,00
	F4	F5	0,09	1,00
		F6	0,57	0,11
		F1	-0,27	1,00
		F2	-0,67	0,02
		F3	-0,18	1,00
		NF1	0,27	1,00
		NF2	1,11	0,00
	NF1	F5	-0,17	1,00
		F6	0,30	1,00
		F1	-0,53	0,18
		F2	-0,94	0,00
		F3	-0,44	0,66
		F4	-0,27	1,00
		NF2	0,85	0,00
	NF2	F5	-1,02	0,00
		F6	-0,54	0,15
		F1	-1,38	0,00
		F2	-1,78	0,00
		F3	-1,29	0,00
		F4	-1,11	0,00
		NF1	-0,85	0,00
Cool	F5	F6	-0,15	1,00
		F1	-0,81	0,00
		F2	-0,63	0,05
		F3	-0,52	0,29
		F4	-0,33	1,00
		NF1	1,38	0,00
		NF2	1,24	0,00
	F6	F5	0,15	1,00
		F1	-0,66	0,02
		F2	-0,48	0,43
		F3	-0,37	1,00
		F4	-0,18	1,00
		NF1	1,53	0,00
		NF2	1,39	0,00
	F1	F5	0,81	0,00
		F6	0,66	0,02
		F2	0,18	1,00
		F3	0,29	1,00
		F4	0,48	0,43
		NF1	2,18	0,00
		NF2	2,05	0,00
	F2	F5	0,63	0,05
		F6	0,48	0,43

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F1	-0,18	1,00
		F3	0,11	1,00
		F4	0,30	1,00
		NF1	2,01	0,00
		NF2	1,87	0,00
	F3	F5	0,52	0,29
		F6	0,37	1,00
		F1	-0,29	1,00
		F2	-0,11	1,00
		F4	0,19	1,00
		NF1	1,89	0,00
		NF2	1,76	0,00
	F4	F5	0,33	1,00
		F6	0,18	1,00
		F1	-0,48	0,43
		F2	-0,30	1,00
		F3	-0,19	1,00
		NF1	1,70	0,00
		NF2	1,57	0,00
	NF1	F5	-1,38	0,00
		F6	-1,53	0,00
		F1	-2,18	0,00
		F2	-2,01	0,00
		F3	-1,89	0,00
		F4	-1,70	0,00
		NF2	-0,14	1,00
	NF2	F5	-1,24	0,00
		F6	-1,39	0,00
		F1	-2,05	0,00
		F2	-1,87	0,00
		F3	-1,76	0,00
		F4	-1,57	0,00
		NF1	0,14	1,00
Calmed	F5	F6	0,37	1,00
		F1	-0,27	1,00
		F2	-0,43	1,00
		F3	0,02	1,00
		F4	-0,29	1,00
		NF1	2,70	0,00
		NF2	2,63	0,00
	F6	F5	-0,37	1,00
		F1	-0,64	0,17
		F2	-0,80	0,02
		F3	-0,35	1,00
		F4	-0,66	0,17
		NF1	2,33	0,00
		NF2	2,26	0,00
	F1	F5	0,27	1,00
		F6	0,64	0,17
		F2	-0,16	1,00
		F3	0,29	1,00
		F4	-0,02	1,00
		NF1	2,97	0,00
		NF2	2,90	0,00
	F2	F5	0,43	1,00
		F6	0,80	0,02
		F1	0,16	1,00
		F3	0,44	1,00
		F4	0,14	1,00
		NF1	3,13	0,00
		NF2	3,06	0,00
	F3	F5	-0,02	1,00
		F6	0,35	1,00
		F1	-0,29	1,00
		F2	-0,44	1,00
		F4	-0,30	1,00
		NF1	2,68	0,00
		NF2	2,61	0,00
	F4	F5	0,29	1,00
		F6	0,66	0,17
		F1	0,02	1,00
		F2	-0,14	1,00
		F3	0,30	1,00
		NF1	2,99	0,00

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		NF2	2,91	0,00
	NF1	F5	-2,70	0,00
		F6	-2,33	0,00
		F1	-2,97	0,00
		F2	-3,13	0,00
		F3	-2,68	0,00
		F4	-2,99	0,00
		NF2	-0,07	1,00
	NF2	F5	-2,63	0,00
		F6	-2,26	0,00
		F1	-2,90	0,00
		F2	-3,06	0,00
		F3	-2,61	0,00
		F4	-2,91	0,00
		NF1	0,07	1,00
Smooth	F5	F6	0,01	1,00
		F1	-0,27	1,00
		F2	-0,73	0,00
		F3	-0,19	1,00
		F4	-0,33	1,00
		NF1	0,16	1,00
		NF2	0,35	1,00
	F6	F5	-0,01	1,00
		F1	-0,28	1,00
		F2	-0,74	0,00
		F3	-0,20	1,00
		F4	-0,34	1,00
		NF1	0,16	1,00
		NF2	0,35	1,00
	F1	F5	0,27	1,00
		F6	0,28	1,00
		F2	-0,46	0,30
		F3	0,08	1,00
		F4	-0,06	1,00
		NF1	0,44	0,44
		NF2	0,63	0,01
	F2	F5	0,73	0,00
		F6	0,74	0,00
		F1	0,46	0,30
		F3	0,54	0,08
		F4	0,40	0,81
		NF1	0,89	0,00
		NF2	1,08	0,00
	F3	F5	0,19	1,00
		F6	0,20	1,00
		F1	-0,08	1,00
		F2	-0,54	0,08
		F4	-0,14	1,00
		NF1	0,35	1,00
		NF2	0,55	0,06
	F4	F5	0,33	1,00
		F6	0,34	1,00
		F1	0,06	1,00
		F2	-0,40	0,81
		F3	0,14	1,00
		NF1	0,49	0,21
		NF2	0,68	0,00
	NF1	F5	-0,16	1,00
		F6	-0,16	1,00
		F1	-0,44	0,44
		F2	-0,89	0,00
		F3	-0,35	1,00
		F4	-0,49	0,21
		NF2	0,19	1,00
	NF2	F5	-0,35	1,00
		F6	-0,35	1,00
		F1	-0,63	0,01
		F2	-1,08	0,00
		F3	-0,55	0,06
		F4	-0,68	0,00
		NF1	-0,19	1,00
Explorer	F5	F6	0,00	1,00
		F1	-0,49	0,72

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F2	-0,27	1,00
		F3	-0,22	1,00
		F4	0,37	1,00
		NF1	-1,40	0,00
		NF2	-0,97	0,00
	F6	F5	0,00	1,00
		F1	-0,49	0,67
		F2	-0,27	1,00
		F3	-0,22	1,00
		F4	0,37	1,00
		NF1	-1,40	0,00
		NF2	-0,97	0,00
	F1	F5	0,49	0,72
		F6	0,49	0,67
		F2	0,22	1,00
		F3	0,27	1,00
		F4	0,86	0,00
		NF1	-0,91	0,00
		NF2	-0,48	0,67
	F2	F5	0,27	1,00
		F6	0,27	1,00
		F1	-0,22	1,00
		F3	0,05	1,00
		F4	0,64	0,10
		NF1	-1,13	0,00
		NF2	-0,70	0,04
	F3	F5	0,22	1,00
		F6	0,22	1,00
		F1	-0,27	1,00
		F2	-0,05	1,00
		F4	0,59	0,20
		NF1	-1,18	0,00
		NF2	-0,75	0,02
	F4	F5	-0,37	1,00
		F6	-0,37	1,00
		F1	-0,86	0,00
		F2	-0,64	0,10
		F3	-0,59	0,20
		NF1	-1,77	0,00
		NF2	-1,34	0,00
	NF1	F5	1,40	0,00
		F6	1,40	0,00
		F1	0,91	0,00
		F2	1,13	0,00
		F3	1,18	0,00
		F4	1,77	0,00
		NF2	0,43	1,00
	NF2	F5	0,97	0,00
		F6	0,97	0,00
		F1	0,48	0,67
		F2	0,70	0,04
		F3	0,75	0,02
		F4	1,34	0,00
		NF1	-0,43	1,00
Light	F5	F6	-0,26	1,00
		F1	-1,06	0,00
		F2	-1,26	0,00
		F3	-0,73	0,02
		F4	-0,87	0,00
		NF1	-0,11	1,00
		NF2	0,71	0,02
	F6	F5	0,26	1,00
		F1	-0,80	0,00
		F2	-1,00	0,00
		F3	-0,46	0,72
		F4	-0,60	0,12
		NF1	0,15	1,00
		NF2	0,97	0,00
	F1	F5	1,06	0,00
		F6	0,80	0,00
		F2	-0,20	1,00
		F3	0,34	1,00
		F4	0,20	1,00
		NF1	0,95	0,00

(*continued on next page*)

Table A1 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		NF2	1,77	0,00
	F2	F5	1,26	0,00
		F6	1,00	0,00
		F1	0,20	1,00
		F3	0,54	0,26
		F4	0,40	1,00
		NF1	1,15	0,00
		NF2	1,97	0,00
	F3	F5	0,73	0,02
		F6	0,46	0,72
		F1	-0,34	1,00
		F2	-0,54	0,26
		F4	-0,14	1,00
		NF1	0,61	0,09
		NF2	1,43	0,00
	F4	F5	0,87	0,00
		F6	0,60	0,12
		F1	-0,20	1,00
		F2	-0,40	1,00
		F3	0,14	1,00
		NF1	0,75	0,01
		NF2	1,57	0,00
	NF1	F5	0,11	1,00
		F6	-0,15	1,00
		F1	-0,95	0,00
		F2	-1,15	0,00
		F3	-0,61	0,09
		F4	-0,75	0,01
		NF2	0,82	0,00
	NF2	F5	-0,71	0,02
		F6	-0,97	0,00
		F1	-1,77	0,00
		F2	-1,97	0,00
		F3	-1,43	0,00
		F4	-1,57	0,00
		NF1	-0,82	0,00

Table A2
Pairwise comparisons between auditory conditions on each olfactory note of Experiment 1.

Dependent Variable	Stimuli		Mean Difference	p-value
Spicy	F5	F6	-0,05	1,00
		F1	0,13	1,00
		F2	0,23	1,00
		F3	0,10	1,00
		F4	0,14	1,00
		NF1	-2,42	0,00
		NF2	-2,43	0,00
	F6	F5	0,05	1,00
		F1	0,18	1,00
		F2	0,28	1,00
		F3	0,15	1,00
		F4	0,19	1,00
		NF1	-2,37	0,00
		NF2	-2,38	0,00
	F1	F5	-0,13	1,00
		F6	-0,18	1,00
		F2	0,09	1,00
		F3	-0,04	1,00
		F4	0,01	1,00
		NF1	-2,55	0,00
		NF2	-2,57	0,00
	F2	F5	-0,23	1,00
		F6	-0,28	1,00
		F1	-0,09	1,00
		F3	-0,13	1,00
		F4	-0,09	1,00
		NF1	-2,65	0,00
		NF2	-2,66	0,00
	F3	F5	-0,10	1,00
		F6	-0,15	1,00
		F1	0,04	1,00

(*continued on next page*)

Table A2 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F2	0,13	1,00
		F4	0,04	1,00
		NF1	-2,52	0,00
		NF2	-2,53	0,00
	F4	F5	-0,14	1,00
		F6	-0,19	1,00
		F1	-0,01	1,00
		F2	0,09	1,00
		F3	-0,04	1,00
		NF1	-2,56	0,00
		NF2	-2,57	0,00
	NF1	F5	2,42	0,00
		F6	2,37	0,00
		F1	2,55	0,00
		F2	2,65	0,00
		F3	2,52	0,00
		F4	2,56	0,00
		NF2	-0,01	1,00
	NF2	F5	2,43	0,00
		F6	2,38	0,00
		F1	2,57	0,00
		F2	2,66	0,00
		F3	2,53	0,00
		F4	2,57	0,00
		NF1	0,01	1,00
Fresh	F5	F6	-0,28	1,00
		F1	-1,40	0,00
		F2	-1,30	0,00
		F3	-0,62	0,83
		F4	-0,37	1,00
		NF1	0,49	1,00
		NF2	0,72	0,31
	F6	F5	0,28	1,00
		F1	-1,12	0,00
		F2	-1,02	0,01
		F3	-0,34	1,00
		F4	-0,09	1,00
		NF1	0,77	0,17
		NF2	1,00	0,01
	F1	F5	1,40	0,00
		F6	1,12	0,00
		F2	0,10	1,00
		F3	0,78	0,13
		F4	1,02	0,01
		NF1	1,89	0,00
		NF2	2,12	0,00
	F2	F5	1,30	0,00
		F6	1,02	0,01
		F1	-0,10	1,00
		F3	0,68	0,40
		F4	0,93	0,03
		NF1	1,79	0,00
		NF2	2,02	0,00
	F3	F5	0,62	0,83
		F6	0,34	1,00
		F1	-0,78	0,13
		F2	-0,68	0,40
		F4	0,24	1,00
		NF1	1,11	0,00
		NF2	1,34	0,00
	F4	F5	0,37	1,00
		F6	0,09	1,00
		F1	-1,02	0,01
		F2	-0,93	0,03
		F3	-0,24	1,00
		NF1	0,87	0,07
		NF2	1,09	0,00
	NF1	F5	-0,49	1,00
		F6	-0,77	0,17
		F1	-1,89	0,00
		F2	-1,79	0,00
		F3	-1,11	0,00
		F4	-0,87	0,07
		NF2	0,23	1,00

(*continued on next page*)

Table A2 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
	NF2	F5	-0,72	0,31
		F6	-1,00	0,01
		F1	-2,12	0,00
		F2	-2,02	0,00
		F3	-1,34	0,00
		F4	-1,09	0,00
		NF1	-0,23	1,00
Fruity	F5	F6	0,22	1,00
		F1	-0,33	1,00
		F2	-1,28	0,00
		F3	-0,42	1,00
		F4	-0,68	0,28
		NF1	-0,33	1,00
		NF2	-0,05	1,00
	F6	F5	-0,22	1,00
		F1	-0,55	0,89
		F2	-1,49	0,00
		F3	-0,64	0,36
		F4	-0,90	0,02
		NF1	-0,55	1,00
		NF2	-0,27	1,00
	F1	F5	0,33	1,00
		F6	0,55	0,89
		F2	-0,95	0,01
		F3	-0,09	1,00
		F4	-0,35	1,00
		NF1	0,00	1,00
		NF2	0,28	1,00
	F2	F5	1,28	0,00
		F6	1,49	0,00
		F1	0,95	0,01
		F3	0,85	0,03
		F4	0,59	0,65
		NF1	0,95	0,01
		NF2	1,23	0,00
	F3	F5	0,42	1,00
		F6	0,64	0,36
		F1	0,09	1,00
		F2	-0,85	0,03
		F4	-0,26	1,00
		NF1	0,09	1,00
		NF2	0,37	1,00
	F4	F5	0,68	0,28
		F6	0,90	0,02
		F1	0,35	1,00
		F2	-0,59	0,65
		F3	0,26	1,00
		NF1	0,36	1,00
		NF2	0,63	0,41
	NF1	F5	0,33	1,00
		F6	0,55	1,00
		F1	0,00	1,00
		F2	-0,95	0,01
		F3	-0,09	1,00
		F4	-0,36	1,00
		NF2	0,28	1,00
	NF2	F5	0,05	1,00
		F6	0,27	1,00
		F1	-0,28	1,00
		F2	-1,23	0,00
		F3	-0,37	1,00
		F4	-0,63	0,41
		NF1	-0,28	1,00
Citrus	F5	F6	-0,23	1,00
		F1	-0,54	0,84
		F2	-0,58	0,57
		F3	-0,23	1,00
		F4	0,15	1,00
		NF1	-0,97	0,00
		NF2	-0,69	0,15
	F6	F5	0,23	1,00
		F1	-0,31	1,00
		F2	-0,35	1,00

(*continued on next page*)

Table A2 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F3	0,00	1,00
		F4	0,38	1,00
		NF1	-0,74	0,08
		NF2	-0,46	1,00
	F1	F5	0,54	0,84
		F6	0,31	1,00
		F2	-0,04	1,00
		F3	0,30	1,00
		F4	0,69	0,14
		NF1	-0,43	1,00
		NF2	-0,16	1,00
	F2	F5	0,58	0,57
		F6	0,35	1,00
		F1	0,04	1,00
		F3	0,35	1,00
		F4	0,73	0,09
		NF1	-0,39	1,00
		NF2	-0,11	1,00
	F3	F5	0,23	1,00
		F6	0,00	1,00
		F1	-0,30	1,00
		F2	-0,35	1,00
		F4	0,39	1,00
		NF1	-0,73	0,08
		NF2	-0,46	1,00
	F4	F5	-0,15	1,00
		F6	-0,38	1,00
		F1	-0,69	0,14
		F2	-0,73	0,09
		F3	-0,39	1,00
		NF1	-1,12	0,00
		NF2	-0,85	0,02
	NF1	F5	0,97	0,00
		F6	0,74	0,08
		F1	0,43	1,00
		F2	0,39	1,00
		F3	0,73	0,08
		F4	1,12	0,00
		NF2	0,27	1,00
	NF2	F5	0,69	0,15
		F6	0,46	1,00
		F1	0,16	1,00
		F2	0,11	1,00
		F3	0,46	1,00
		F4	0,85	0,02
		NF1	-0,27	1,00
Sweet	F5	F6	0,23	1,00
		F1	0,01	1,00
		F2	-1,27	0,00
		F3	-0,35	1,00
		F4	-0,59	0,98
		NF1	-0,02	1,00
		NF2	-0,10	1,00
	F6	F5	-0,23	1,00
		F1	-0,22	1,00
		F2	-1,49	0,00
		F3	-0,58	0,89
		F4	-0,81	0,08
		NF1	-0,25	1,00
		NF2	-0,33	1,00
	F1	F5	-0,01	1,00
		F6	0,22	1,00
		F2	-1,28	0,00
		F3	-0,36	1,00
		F4	-0,60	0,75
		NF1	-0,03	1,00
		NF2	-0,12	1,00
	F2	F5	1,27	0,00
		F6	1,49	0,00
		F1	1,28	0,00
		F3	0,91	0,02
		F4	0,68	0,36
		NF1	1,24	0,00
		NF2	1,16	0,00

(*continued on next page*)

Table A2 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
	F3	F5	0,35	1,00
		F6	0,58	0,89
		F1	0,36	1,00
		F2	-0,91	0,02
		F4	-0,23	1,00
		NF1	0,33	1,00
		NF2	0,25	1,00
	F4	F5	0,59	0,98
		F6	0,81	0,08
		F1	0,60	0,75
		F2	-0,68	0,36
		F3	0,23	1,00
		NF1	0,56	1,00
		NF2	0,48	1,00
	NF1	F5	0,02	1,00
		F6	0,25	1,00
		F1	0,03	1,00
		F2	-1,24	0,00
		F3	-0,33	1,00
		F4	-0,56	1,00
		NF2	-0,08	1,00
	NF2	F5	0,10	1,00
		F6	0,33	1,00
		F1	0,12	1,00
		F2	-1,16	0,00
		F3	-0,25	1,00
		F4	-0,48	1,00
		NF1	0,08	1,00
Spices	F5	F6	0,11	1,00
		F1	0,12	1,00
		F2	-0,09	1,00
		F3	0,31	1,00
		F4	0,09	1,00
		NF1	-0,62	0,48
		NF2	-0,78	0,07
	F6	F5	-0,11	1,00
		F1	0,01	1,00
		F2	-0,20	1,00
		F3	0,21	1,00
		F4	-0,02	1,00
		NF1	-0,73	0,13
		NF2	-0,89	0,01
	F1	F5	-0,12	1,00
		F6	-0,01	1,00
		F2	-0,21	1,00
		F3	0,19	1,00
		F4	-0,03	1,00
		NF1	-0,74	0,10
		NF2	-0,90	0,01
	F2	F5	0,09	1,00
		F6	0,20	1,00
		F1	0,21	1,00
		F3	0,41	1,00
		F4	0,18	1,00
		NF1	-0,53	1,00
		NF2	-0,69	0,18
	F3	F5	-0,31	1,00
		F6	-0,21	1,00
		F1	-0,19	1,00
		F2	-0,41	1,00
		F4	-0,22	1,00
		NF1	-0,93	0,01
		NF2	-1,09	0,00
	F4	F5	-0,09	1,00
		F6	0,02	1,00
		F1	0,03	1,00
		F2	-0,18	1,00
		F3	0,22	1,00
		NF1	-0,71	0,16
		NF2	-0,87	0,02
	NF1	F5	0,62	0,48
		F6	0,73	0,13
		F1	0,74	0,10
		F2	0,53	1,00

(*continued on next page*)

Table A2 (*continued*)

Dependent Variable	Stimuli		Mean Difference	p-value
		F3	0,93	0,01
		F4	0,71	0,16
		NF2	-0,16	1,00
	NF2	F5	0,78	0,07
		F6	0,89	0,01
		F1	0,90	0,01
		F2	0,69	0,18
		F3	1,09	0,00
		F4	0,87	0,02
		NF1	0,16	1,00

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